Local–edge routing for conformational B-cell epitope prediction: what survives leakage-free selection

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Abstract

Meta-predictors are common in conformational B-cell epitope prediction: a mixing weight, an ensemble coefficient or a voting rule is tuned, and the combined score is reported as an improvement. When that parameter is chosen on the data the improvement is reported on, the improvement is partly the choosing, and we quantify how much. The case we audit is one of our own. An earlier unpublished analysis had reported that fusing a local predictor with BepiPred-3.0 improves on BepiPred-3.0 alone; reproducing it to within 4.9e-07 across all 44 entries of its stored parameter grid, we show that both tuning parameters, the mixing weight and the decision threshold, had been selected on the evaluation split. Under leave-one-sequence-out nested cross-validation the advantage is no longer detectable (ΔROC-AUC -0.0005 [-0.0033, +0.0021]) and the MCC estimate changes sign, from +0.0115 to -0.0329. Resampling turns that single instance into a quantity: we draw antigen subsets and estimate the fused-minus-reference difference twice on each, once by scanning the parameter grid on the subset and once by nested cross-validation within it. Model selection bias, the gap between the two, contributes 0.0542 MCC at 4 antigens and 0.0315 at 12; it rises across the grid densities examined and, at the densities ordinarily used, is positive in 100% of draws. Because that inflation exceeds the +0.0115 the tuned protocol had reported, a gain of that size is uninformative here. Nor is the practice ours alone: of 12 published combined epitope predictors, 3 state that the combination parameter was chosen on the evaluation data, 6 state a separate basis, and 3 do not say. The same harness measures deployment cost. Routing antigens to a remote model on the local model's uncertainty runs at 13.6 ms against 42.4 ms for always-remote execution ([-37.9, -19.1]) and transmits 0.285 of raw sequences, while its quality effect is not resolved at 15 antigens. Modelled carbon favours routing only above a ratio: routing is the most carbon-intensive policy for the pair measured here, but the break-even rises with remote model size, and the ordering reverses above 13.0× the measured remote operational cost, which BepiPred-3.0's own backbone exceeds. Throughout, intervals over antigens are an order of magnitude wider than over re-runs; on benchmarks of this size the conventional analyses report more confidence than the data supports.

Keywords: conformational B-cell epitope prediction; edge inference; model selection; data leakage; reproducible benchmarking

1 Introduction

A laboratory that has just expressed a new antigen and wants to know which of its residues belong to conformational B-cell epitopes has good computational options, and a practical problem in using them. The best current conformational predictors are built on protein language models whose weights run to billions of parameters (Lin et al., 2023). Running one well needs either a workstation with a capable accelerator or, far more commonly, uploading the sequence to a hosted

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service. Neither is comfortable for every group, and accelerators are unevenly distributed. For antigens from unpublished constructs, from clinical isolates, or from an industrial pipeline, the sequence itself is the sensitive object: sending it to a third-party endpoint is a disclosure decision, not a technical one.

The obvious middle path is to keep a small model on the local machine and call the remote service only when the local model is unsure. Systems research has established the idea under names such as split computing, model cascades and early exiting (Matsubara et al., 2022). Its appeal in a screening setting is easy to state: most antigens should be easy, the easy ones never leave the device, and the hard minority gets the benefit of the larger model. What such a scheme costs and returns is much less clear. A laboratory cares about three quantities at once: how good the predictions are, how long screening takes, and how much sequence leaves the building. We measure all three on the same runs rather than arguing them separately.

Quality is the first of the three, and it is the one most easily overstated. The paper re-examines an earlier unpublished analysis of our own, which had reported that score fusion, blending the per-residue scores of our local predictor with those of BepiPred-3.0 (Clifford et al., 2022), improved on BepiPred-3.0 alone. Two tuning parameters govern that blend, the mixing weight and the decision threshold, and the earlier analysis had fixed both by scoring a parameter grid on the evaluation split and keeping the best-scoring row. Kapoor and Narayanan (2023) identify such optimization of thresholds on evaluation data as the most common form of leakage in applied machine learning. The step is easy to commit, because a sensitivity analysis and a parameter selection procedure look superficially alike. We reproduce the original numbers under the original protocol, to within 4.9e-07. We then re-estimate the same quantity under nested cross-validation, with the weight and the threshold selected in an inner loop over the training folds while antigens are held out one at a time in the outer loop. The advantage does not survive on either dataset. On BP3C50ID the difference against BepiPred-3.0 is no longer detectable and the MCC estimate changes sign. On a larger IEDB-derived subset, which fuses against our own local predictor in place of BepiPred-3.0, the difference is likewise indistinguishable from zero. On both, the inner loop puts almost all of the weight on the reference predictor and almost none on ours. Neither dataset is large enough to show that fusion cannot help, and we do not claim that it does not.

The literature this belongs to is accumulating fusion and ensemble results faster than it scrutinises how their tuning parameters were chosen. The corrective is cheap, since the nested protocol here costs one extra outer loop over antigens and needs no new data.

What a routed deployment costs is the remaining question, and it needs the same discipline. We build a measurement harness that runs a set of routing policies, from always-local through uncertainty-triggered routing to always-remote, on identical inputs with identical seeds. It records predictive quality, latency, data movement and an energy model per antigen. Over 10 seeds on the BP3C50ID external split, the answer arrives with intervals attached. Routing on local uncertainty gives up nothing detectable in quality against always-remote execution, cuts latency to roughly a third, and keeps most raw sequences on the device. It also gives an answer nobody in this project expected. A routed antigen is charged twice: once for the local pass that decides it is hard, once for the remote pass that then processes it. For the checkpoint pair measured here, routing is therefore the most carbon-intensive policy in the comparison. The green framing that usually accompanies edge inference does not survive contact with the accounting. The condition under which it would is a ratio rather than a verdict. Section 4.6 shows that a remote tier the size of the one a laboratory would really call sits on the other side of it.

Contributions. This paper offers the following.

1. A reproducible measurement harness for conformational B-cell epitope screening. It records predictive quality, latency, data movement and modelled energy together, per antigen and per seed, and states the provenance of each cost quantity (Section 3).

2. An instance, in conformational epitope prediction, of the model selection bias documented

in bioinformatics since Varma and Simon (2006). We reproduce a previously reported fusion advantage under its original, test-derived protocol, then re-estimate it under nested cross-validation on two datasets. The advantage is no longer detectable on either, and on BP3C50ID the estimate changes sign (Section 4.2).

3. The observation that, given a free choice on either dataset, the inner loop assigns our predictor almost no weight. We report the per-fold selections rather than summarising them away (Section 4.3).

4. A survey of 12 published B-cell epitope predictors that combine two or more models, the consensus predictors and meta-predictors of this field, asking of each whether the combination parameter was selected on the data the headline result is reported on. 3 state that it was, 6 state a separate basis, and 3 do not say. Each judgement is recorded with the verbatim sentence it rests on (Section 4.4).

5. A resampling estimate of how large the model selection bias is, as a function of how much freedom the parameter grid allows. The inflation runs from 0.0542 MCC at 4 antigens to 0.0315 at 12, and rises across the densities of the weight grid examined: at 12 antigens a 3-point grid yields 0.0184 and a 41-point grid 0.0290. At the grid densities ordinarily used the test-derived estimate was the more favourable one in 100% of draws (Section 4.5).

6. A 10-seed characterisation of five local–edge routing policies on BP3C50ID, with bootstrap intervals and paired sign tests. It includes a closed-form condition locating where the known offloading energy threshold (Kumar and Lu, 2010) falls for this task, and how that location moves with the size of the remote tier. Routing increases modelled carbon once the routed fraction exceeds 0.139 for the checkpoint pair measured here; the ordering reverses above a remote operational cost of 13.0× the one measured, which the reference predictor's own backbone exceeds (Section 4.6).

What the two halves give together. Figure 1 puts the deployment and its evaluation on one page, and the pairing is what this work contributes. Panel (a) is a measurement a laboratory can act on: for one antigen, the same runs yield the latency, the share of sequence that leaves the device and the modelled carbon of each routing policy, so the three can be weighed against one another instead of argued separately. Panel (b) decides whether the quality column of that readout means anything. Running both selection protocols over the same antigens turns a disagreement about method into a measured quantity, and that quantity is larger than the effect it was invoked to support. Two things follow that a single-protocol study cannot reach. The size of the selection step can be tracked as the antigen count and the grid density vary, and separated from the difference between pooling residues and averaging over antigens. And the cost side can be pushed past the hardware actually measured: panel (c) locates the remote scale at which the carbon ordering reverses, which the reference predictor's own backbone already exceeds.

2 Background

2.1 Conformational B-cell epitope prediction

A B-cell epitope is the part of an antigen that an antibody binds in an antibody–antigen complex. Linear B-cell epitopes are contiguous in sequence; conformational, or discontinuous, epitopes, which are the large majority of what antibodies actually recognise, are assembled from residues that the folded structure brings together but that may be far apart in the chain. Prediction is therefore per-residue binary labelling, each residue being called an epitope residue or not, and two properties of that task are awkward. Epitope residues are a minority of all residues, and the label depends on structural context that the sequence encodes only implicitly.

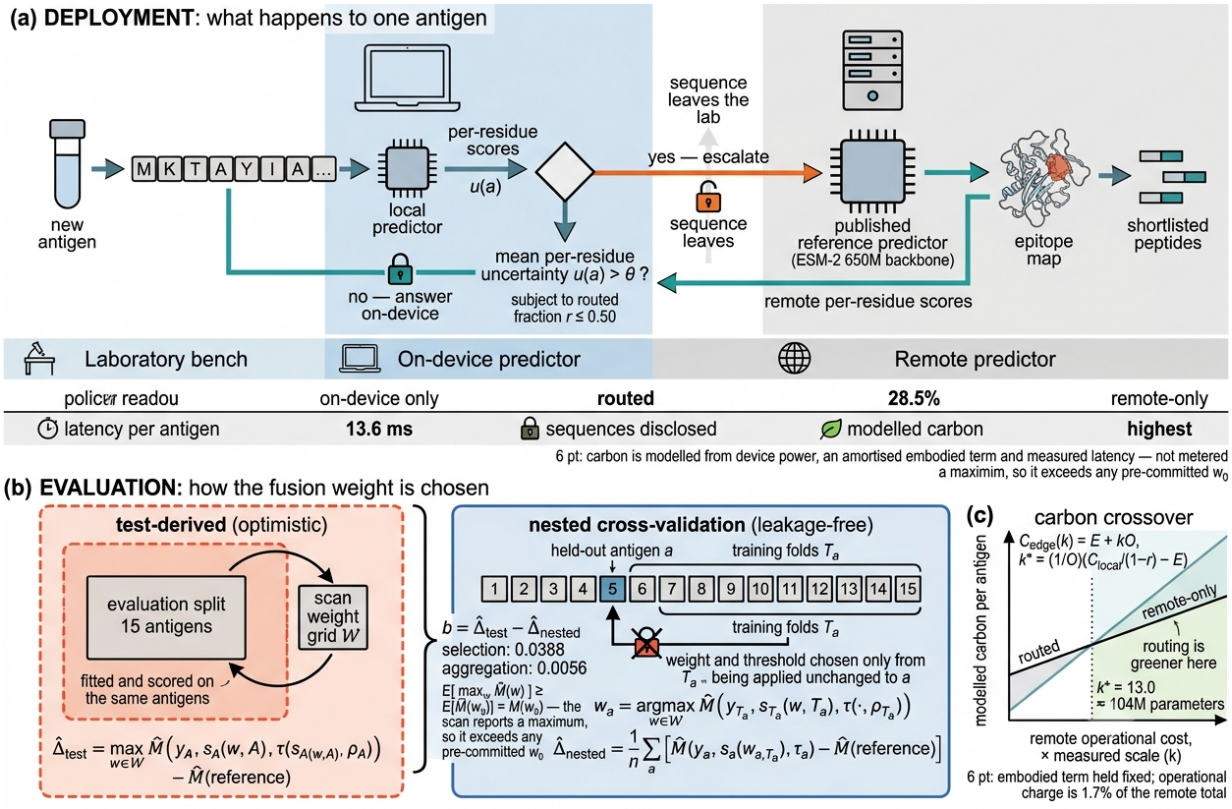


Figure 1: Routing antigens between a bench predictor and a published remote model, and the protocol used to evaluate it. **(a)** For each newly expressed antigen the on-device predictor runs first; antigens whose mean per-residue uncertainty exceeds a fitted threshold are escalated to the remote model, subject to a routed-fraction budget. Escalation is what discloses the sequence, so the padlock opens only on the uplink. The strip beneath compares the routed policy with always-local and always-remote execution on latency, disclosure and modelled carbon. **(b)** The mixing weight and the decision threshold can be chosen two ways. The test-derived protocol scans the weight grid on the same antigens it reports, so its arrow returns into its own source; nested cross-validation chooses both on the training folds alone and applies them unchanged to the held-out antigen. **(c)** Modelled carbon favours always-remote execution only below a crossover in remote operational cost. Carbon throughout is modelled from device power, an amortised embodied term and measured latency, and is not metered.

Methods have moved through three broad generations. Propensity-scale approaches scored residues with physicochemical heuristics. Structure-based methods used solved or modelled structures to compute relative solvent accessibility and protrusion for each surface residue. The current generation feeds representations from a pre-trained protein language model into a supervised head, which is how BepiPred-3.0 (Clifford et al., 2022) and structure-aware successors such as DiscoTope-3.0 (Høie et al., 2024) are built. Recent additions include the SEMA 2.0 web platform (Ivanisenko et al., 2024) and residue-level predictors built directly on ESM-2 projections (Israeli and Louzoun, 2024). Both sit close to the PLM family evaluated here. Gabellieri et al. (2026) survey the field as it now stands. We use BepiPred-3.0 as the reference predictor on BP3C50ID, where it is the strongest openly runnable sequence-based baseline and the method a laboratory would realistically be choosing between. On the IEDB-derived subset we do not have BepiPred-3.0 predictions, so the reference there is our own local-only predictor. The two fusion arms therefore test different pairs, and we keep them separate throughout.

Evaluation in this area is dominated by the non-redundant BP3C50ID benchmark distributed with BepiPred-3.0, whose external test split holds 15 antigens (2670 residues, 507 of them positive). That independent benchmark is a small one, and its size is worth keeping in view when reading any comparison built on it, including ours. We report the Matthews correlation coefficient (MCC) alongside the threshold-independent ROC-AUC and average precision, because MCC behaves

sensibly under the class imbalance that per-residue epitope labelling produces (Chicco and Jurman, 2020). Under that imbalance, accuracy and F1 can both be high for a predictor that has learned little.

2.2 Splitting inference between a local device and a remote service

Running part of a model locally and part remotely is an old idea with a current name. The systems literature treats it as a scheduling problem. The task is to decide where each input is processed, given a local model, a remote model, and a per-input signal about difficulty, so that some combination of accuracy, latency, cost and bandwidth is optimised (Matsubara et al., 2022). The signal is usually the local model's own confidence. That makes the mechanism attractively simple, and it also introduces its characteristic failure, since the decision is only as good as the calibration of the model making it.

Whether offloading saves energy at all is a settled question in that literature. The answer depends on the ratio of computation offloaded to the overhead of offloading it. Below a threshold ratio, offloading costs more energy than it saves (Kumar and Lu, 2010). The same effect has been observed recently in model cascades, where a draft-and-verify decoding scheme consumed more energy than running the large model alone despite reducing latency (Dutta et al., 2026). Our contribution on this axis is not the phenomenon but the location of the threshold for this task, and the observation that per-residue epitope screening sits on the unfavourable side of it.

Two things are different in the setting we study. The first is that the quantity a laboratory wants to minimise is often not bandwidth or cost but disclosure, meaning how many of its sequences are transmitted at all. Disclosure is a counting measure rather than a continuous one, and no accuracy-oriented objective captures it. The second is that the per-antigen compute is small enough that the routing decision is not free relative to the work it schedules. In large-model serving, a confidence check is negligible beside the inference it gates. In per-residue epitope screening on short sequences, it is not. The carbon results in Section 4.6 are a consequence of that ratio.

2.3 Accounting for energy and carbon

Reported carbon figures for machine-learning systems are rarely measurements. The common practice, which we follow and label as such, is to derive energy from a device power figure and a measured runtime. That energy is then converted to mass of carbon dioxide equivalent with a grid-intensity constant (Strubell et al., 2019; Lacoste et al., 2019; Patterson et al., 2021). Our energy model follows the formulation of Green Algorithms (Lannelongue et al., 2021), which also supplies the amortised-hardware term. That formulation has been applied to bioinformatics workloads specifically (Grealey et al., 2022). We claim no novelty in the instrument. The result is a model whose output is proportional to measured latency by construction. The model is useful for comparing policies that run on the same hardware, since the proportionality constant cancels; it is not a measurement of what any particular laboratory's equipment would draw, and we are careful not to present it as one. Table 5 records which of our reported quantities are measured and which are modelled.

2.4 Selection on the evaluation split

When a method has a tuning parameter, an ensemble weight or a decision threshold or a fusion coefficient, and that parameter is chosen by scoring a parameter grid on the data the paper reports, the reported figure is an over-optimistic estimate of the predictor's true error. Model selection bias of this kind grows with the number of grid points tried and shrinks with the size of the evaluation set, and it is not a new observation.

The mechanism itself is one line of algebra, and it is worth setting down because it fixes what can and cannot be concluded from it. Write $\mathcal{W}_G$ for a grid of G candidate values of the tuning parameter, $\widehat{M}(w)$ for the metric of the configuration indexed by w computed on the split

the figure is reported on, and $M(w)$ for the quantity that $\widehat{M}(w)$ estimates. A grid scan reports $\max_{w \in \mathcal{W}_G} \widehat{M}(w)$. For any grid point $w_0 \in \mathcal{W}_G$ fixed in advance of seeing that split, and for any finer grid $\mathcal{W}_{G'} \supseteq \mathcal{W}_G$,

$$\mathbb{E}\Big[\max_{w \in \mathcal{W}_G} \widehat{M}(w)\Big] \;\geq\; \mathbb{E}\big[\widehat{M}(w_0)\big] \;=\; M(w_0), \qquad \mathbb{E}\Big[\max_{w \in \mathcal{W}_G} \widehat{M}(w)\Big] \;\leq\; \mathbb{E}\Big[\max_{w \in \mathcal{W}_{G'}} \widehat{M}(w)\Big]. \tag{1}$$

Both parts hold split by split before any expectation is taken: the maximum of a finite set is at least as large as any member of it, which gives the first inequality, and enlarging the set over which a maximum is taken cannot decrease it, which gives the second; expectation preserves both orderings. The single equality is the one assumption, namely that at a grid point chosen without reference to the split, $\widehat{M}(w_0)$ is unbiased for $M(w_0)$, which is what a held-out split buys and what the maximum destroys.

Equation (1) settles two things and no more. It fixes the sign: a scanned maximum sits on average at or above the performance of any configuration one could have committed to beforehand, so the direction of the distortion is known before any data are seen. And it makes the effect non-decreasing under refinement of a *nested* family of grids, which is why a wider grid is the more flattering protocol. That condition is worth stating, because equally spaced grids of different sizes are not nested: the weight 0.25 lies in a five-point grid but not in an eleven-point one, so two such grids are not ordered by inclusion and the inequality says nothing about that pair. It supplies no magnitude either. How far above $M(w_0)$ the expected maximum sits depends on the correlation and the noise scale of the $\widehat{M}(w)$ across the grid, which the inequality does not constrain; the bound is not tight, and it holds with equality only where the scan returns w_0, or a grid point with the same $\widehat{M}$, on every split. The magnitude reported in this paper is therefore measured and not derived: Section 4.5 estimates it by resampling, and that measured gap is an estimate of the difference between two protocols on these data, not of the expected bias in Equation (1).

Varma and Simon (2006) demonstrated it for bioinformatics classifiers two decades ago and showed that nested resampling, which estimates the error in an outer loop over the left-out samples, returns a nearly unbiased estimate. Boulesteix (2009) named the resulting pattern over-optimism. Jelizarow et al. (2010) showed that re-evaluating a published positive result without its optimistic selection step can remove the advantage entirely. Kapoor and Narayanan (2023) survey how widely the problem persists across applied machine-learning research and how often it is invisible in the write-up. The cause is that a table of candidate values reads as a sensitivity analysis whether or not the reported configuration was picked from it. The distinction that matters is not whether the parameter grid is shown but whether the reported configuration was chosen using the labels it is then scored against. Section 3.6 sets out the two protocols we compare precisely enough for the difference to be checked.

3 Materials and methods

3.1 Datasets

The primary benchmark is BP3C50ID, the non-redundant split distributed with BepiPred-3.0, in which training and external test antigens are separated at a 50% sequence identity threshold. The external test split contains 15 antigens, 2670 residues and 507 positive residues, an epitope-residue rate just under one residue in five. All BP3C50ID results below are on this independent benchmark; no model or policy parameter is fitted on it.

A second evaluation uses a subset derived from the Immune Epitope Database (Vita et al., 2024), holding 47 test antigens (14408 residues, 2169 positive) alongside 205 training antigens. The subset is drawn from the `IEDBSeqsNotSharedAt20ID.fasta` collection distributed alongside BP3C50ID, which holds 3560 IEDB sequences selected not to share more than 20% sequence identity with the BP3 data, and is then restricted to sequences of 45–595 residues so that the full policy grid remains runnable on a workstation. Every sequence in the subset is present verbatim in that source collection.

Its purpose is to test whether conclusions drawn from 15 antigens survive on a different and larger collection, which requires the two to be genuinely independent. Independence holds by construction through the identity filter. We also tested it directly against all 358 BP3C50ID sequences and found 0 shared identifiers, 0 shared sequences, and a maximum 10-mer containment of 0.000. No antigen in the subset therefore shares even a single 10-residue run with any BP3C50ID antigen.

3.2 Predictors

We evaluate two families of local and remote predictor of conformational B-cell epitopes, and hold the distinction between them fixed across every policy.

The *heuristic family* uses handcrafted per-residue sequence-context features. Its local member is a lightweight residue classifier; its remote member is an extremely randomised trees refiner that also blends the local score and applies sequence smoothing, with the blend weight and smoothing parameters chosen on a validation split.

The *PLM family* replaces the features with frozen embeddings from ESM-2 (Lin et al., 2023), specifically the `esm2_t6_8M_UR50D` checkpoint. The local member places a logistic residue head on those embeddings; the remote member adds contact-derived features and the same trees refiner. A region-constrained variant post-processes the remote probabilities into contiguous regions using contact density, with the region parameters selected on validation data. We use the 8M-parameter checkpoint because it is what runs on the hardware available for this study. That checkpoint is the smallest of the released ESM-2 models. The PLM results here should therefore be read as a lower bound on what the family can do, not as a characterisation of language-model-based epitope prediction in general.

3.3 Routing policies

A routing policy decides, per antigen, whether the remote model is invoked. Five are compared in each family.

local-only The local model answers every antigen. No sequence is transmitted.

edge-only Every antigen is sent to the remote model.

adaptive-split The local model runs first; if its mean per-residue uncertainty exceeds a threshold, the antigen is re-processed remotely and the remote prediction is used. The threshold and a routing budget are selected on the validation split from a grid of 91 candidate values, subject to a maximum routed fraction of 0.50.

carbon-guard As adaptive-split, with an additional constraint that suppresses routing when the modelled carbon cost of the remote call is not justified by the expected quality gain.

conformal-guard (PLM family) As adaptive-split, with a conformal criterion (Angelopoulos and Bates, 2021) replacing the raw uncertainty threshold.

Every selectable parameter in this section is chosen on a validation split carved out of training data at a fraction of 0.20, grouped by sequence so that no antigen contributes residues to both sides. No routing parameter sees the external test split.

3.4 Cost accounting

Four cost quantities are recorded per antigen, and they differ in how directly they are observed; Table 5 states this for each.

Latency is measured, as wall-clock model-prediction time on the executing device. Bytes transferred are computed from the raw sequence length. Raw-sequence share rate is a count: the

fraction of test antigens whose sequence is transmitted at all, which is the quantity a laboratory worried about disclosure would actually audit.

Energy and carbon are modelled, not metered. Operational energy is the product of a fixed device power figure and the measured latency; operational carbon converts that with a grid-intensity constant of $708\,\mathrm{g\,kW^{-1}\,h}$; embodied carbon is a per-request amortisation of a device lifetime figure. The local device is modelled at 34 W and the remote at 145 W.

The embodied term also carries a device-utilisation fraction. We apply it so that a request is charged only the share of the device's lifetime footprint it actually occupies: 0.20 locally and 0.65 remotely. The two fractions reduce the per-request embodied charge from 0.6000 to 0.1200 gCO_2e locally and from 0.2125 to 0.1381 remotely. We state them explicitly because they are the single largest determinant of every carbon figure in this paper. As Section 6 discusses, the comparison between policies is not fully insensitive to them.

The balance between those two terms matters for reading the results, so we state it plainly. At the latencies observed here the operational term is a very small part of each total: a local pass contributes about 7.0e-06 gCO_2e and a remote pass about 2.4e-03, against per-request totals near 0.120 and 0.139. The modelled carbon of a policy is therefore governed almost entirely by how many device-requests it charges, through the amortised embodied term, and only marginally by how long those requests take. Carbon differences between policies here are differences in request counts, not in compute. We make no claim about metered power anywhere in this paper.

3.5 Evaluation protocol for routing policies

The full pipeline (validation split, parameter selection, training and external-split evaluation) is repeated for 10 seeds (42–51).

Repeating a pipeline over seeds on a fixed test split answers a narrow question: how much a number moves when the same antigens are scored again. It says nothing about how the number would move on different antigens, which is what a reader wants to know. Our primary analysis therefore resamples both. For each of 2000 replicates we draw 15 antigens with replacement and 10 seeds with replacement, recompute every policy's pooled residue-level metrics on that resampled set. We take the 2.5th and 97.5th percentiles of the resulting distribution as a 95% interval. One antigen draw is shared across all policies within a replicate, so paired differences stay paired. Intervals are percentile intervals, with no bias correction or studentisation, and the resampling is driven by a fixed seed recorded in the released artifact.

We report seed-only intervals alongside these in Section 4 only to show how different the two are; no claim in this paper rests on them, and the exact sign test over seeds that an earlier version of this work had relied on is not used.

Two cautions apply to how we read these intervals. First, with 15 antigens a percentile cluster bootstrap is known to under-cover for non-linear statistics such as MCC and ROC-AUC, so the intervals should be read as indicative rather than exact. Second, several of the quantities we report are not random in sign. A policy that routes only some antigens cannot be slower than one that routes all of them, and cannot transmit more sequences. For latency and disclosure the interval therefore describes the magnitude of a difference whose direction is fixed by construction. Reporting that such an interval excludes zero would assert nothing. We say so where it applies, and we reserve the language of detection for the quality metrics, where the sign is genuinely in question. There we treat an interval containing zero as no detected difference, and say so rather than describing such a comparison as a trend.

3.6 Two protocols for selecting a fusion weight

Score fusion combines the reference predictor's per-residue score with ours (CARBON-BCE-R, the local-only predictor) into a meta-predictor, a convex blend governed by a mixing weight w. The two predictors are calibrated differently, so a blend of their raw outputs would be dominated by whichever has the wider spread. We therefore map each onto $[0, 1]$ through its own empirical

distribution function before blending. We estimate that distribution on the training folds and apply it to the held-out fold, like any other fitted transform. The mapping is not a rank transform computed over the whole split, which would let held-out residues influence their own normalisation. One consequence is worth stating, because it is easy to assume otherwise. A transform fitted on training data is not exactly rank-preserving on held-out data, so a fold's ROC-AUC after mapping can differ slightly from the same predictor's raw ROC-AUC. The effect is small, under 10^{-3} per fold in our runs, but it is not zero. For this reason we report per-fold paired differences rather than pooled ranking metrics. A decision threshold then converts blended scores into calls. Both w and the threshold are tuning parameters. The whole question is where they come from.

Notation. Let $\mathcal{A}$ be the set of n test antigens, R_a the residues of antigen a, and $y_{a,i} \in \{0,1\}$ the epitope label of residue $i \in R_a$. Each residue carries two raw scores: $r_{a,i}$ from the reference predictor and $c_{a,i}$ from ours. For a set of antigens $T \subseteq \mathcal{A}$, let $\hat{F}_T^{\mathrm{ref}}$ and $\hat{F}_T^{\mathrm{cand}}$ be the empirical distribution functions of the two raw score sets over the residues of T alone. The blended score of residue i of antigen a at weight w, normalised through T, is

$$s_{a,i}(w,T) \;=\; w\,\hat{F}_T^{\mathrm{ref}}\big(r_{a,i}\big) \;+\; (1-w)\,\hat{F}_T^{\mathrm{cand}}\big(c_{a,i}\big), \qquad w \in \mathcal{W} \subset [0,1], \tag{2}$$

so that w is the weight on the *reference* predictor and $1-w$ the weight on ours; $\mathcal{W}$ is the finite grid of candidate weights that is scanned. We write $s_T(w,U)$ for the vector of $s_{a,i}(w,U)$ over the residues of all antigens in T, and y_T for the corresponding labels.

The threshold is not a free scalar either. For a score vector s and a target positive-call rate ρ,

$$\tau(s,\rho) \;=\; s_{(k)}, \qquad k = \mathrm{round}\big(\rho\,|s|\big), \tag{3}$$

the k-th largest entry of s: the operating point is a constrained quantile, chosen so the blend emits the same number of positive calls as the reference. We write ρ_T for the positive-call rate the reference predictor produces at its own published threshold on the residues of T, and $\widehat{M}(y,s,\tau)$ for the empirical selection metric of scores s against labels y at operating point τ. The two protocols below use exactly these objects and differ only in which data the maximisation in w, the distribution functions, and the threshold are allowed to see.

Test-derived selection. The protocol used in the earlier version of this work optimizes both tuning parameters on the evaluation split, reporting

$$\widehat{\Delta}_{\mathrm{test}} = \max_{w\in\mathcal{W}} \widehat{M}\Big(y_{\mathcal{A}},\, s_{\mathcal{A}}(w,\mathcal{A}),\, \tau\big(s_{\mathcal{A}}(w,\mathcal{A}),\rho_{\mathcal{A}}\big)\Big) \;-\; \widehat{M}\Big(y_{\mathcal{A}},\, \hat{F}_{\mathcal{A}}^{\mathrm{ref}}(r),\, \tau\big(\hat{F}_{\mathcal{A}}^{\mathrm{ref}}(r),\rho_{\mathcal{A}}\big)\Big). \tag{4}$$

Every argument on the right-hand side is the whole evaluation split: the grid maximum, the normalisation and the threshold all see $y_{\mathcal{A}}$. Both tuning parameters are thus functions of the test labels, and the margin the protocol reports over the stronger single predictor is an over-optimistic estimate.

Nested selection. The corrected protocol is nested cross-validation with the antigen as the unit of resampling. The outer loop leaves out one antigen at a time, so that $T_a = \mathcal{A} \setminus \{a\}$ carries every quantity fitted for fold a:

$$w_a = \arg\max_{w\in\mathcal{W}} \widehat{M}\Big(y_{T_a},\, s_{T_a}(w,T_a),\, \tau\big(s_{T_a}(w,T_a),\rho_{T_a}\big)\Big), \tag{5}$$

$$\tau_a = \tau\big(s_{T_a}(w_a,T_a),\,\rho_{T_a}\big). \tag{6}$$

The inner loop (5) scores the same parameter grid on the training folds of that fold; it is a grid scan over those folds, evaluated on the very residues it is fitted on, not a further resampling

split inside them. The weight and the threshold (6) are then applied, unchanged, to the held-out antigen, and the outer estimate is the mean of the per-antigen paired differences,

$$\widehat{\Delta}_{\text{nested}} = \frac{1}{n} \sum_{a \in \mathcal{A}} \Big[\widehat{M}\big(y_a,\, s_a(w_a, T_a),\, \tau_a\big) - \widehat{M}\Big(y_a,\, \hat{F}^{\text{ref}}_{T_a}(r_a),\, \tau\big(\hat{F}^{\text{ref}}_{T_a}(r_{T_a}), \rho_{T_a}\big)\Big)\Big]. \tag{7}$$

Comparing (7) with (4): the metric, the blend and the threshold rule (3) are the same functions; only the subscript on the data they are evaluated with has changed. Nothing derived from the held-out antigen enters the configuration applied to it: not its labels, not its score distribution, not its own quantiles. Independence of that kind is what makes the resulting predictor deployable, since it needs no knowledge of a new antigen before scoring its first residue. We report $\widehat{\Delta}_{\text{nested}}$ with a 95% bootstrap interval over antigens, a Wilcoxon signed-rank test and an exact sign test. We also report the weight w_a chosen in every fold, not its median alone.

What nested selection does and does not buy. The property that holds is an independence one. *Claim:* given the data of T_a, the out-of-fold scores and calls for antigen a in (7) are fixed functions of that antigen's own inputs, and in particular do not depend on y_a. *Proof:* by (5) and (6), w_a, τ_a, $\hat{F}^{\text{ref}}_{T_a}$ and $\hat{F}^{\text{cand}}_{T_a}$ are functions of (r, c, y) restricted to T_a; substituting them into (2) makes $s_{a,i}(w_a, T_a)$ a function of $(r_{a,i}, c_{a,i})$ and those four quantities only, and the call is $\mathbf{1}[s_{a,i}(w_a, T_a) \geq \tau_a]$. So y_a enters (7) only through M, as the labels that already-fixed predictions are scored against. The single assumption is the one the design enforces: folds are defined by antigen, so no residue of a belongs to T_a.

What that licenses is reading (7) as an estimate of the performance of the selection *procedure* on an antigen it has not seen. What it does not license is calling $\widehat{\Delta}_{\text{nested}}$ an unbiased estimate of that performance, and we do not claim it is, for two reasons. First, the inner selection (5) is resubstitution: the metric is maximised on the same training residues at the threshold fitted on those residues, so w_a is itself the argmax of a noisy in-sample score, and (7) inherits whatever optimism that choice carries into the held-out fold. Second, any two folds share $n - 2$ antigens, so the per-antigen differences in (7) are not independent; the spread of the leave-one-out mean is smaller than independent folds would give, which is why the percentile interval over folds undercovers rather than attaining its nominal level. The nested number is the better of the two estimates on offer, not a guaranteed-unbiased one.

Both protocols are implemented in the same module against the same loaders and metric code, so that a difference between their outputs is attributable to the protocol rather than to an implementation difference. Section 4.1 verifies this by reproducing the earlier figures under the test-derived protocol, to within 4.9e-07.

4 Results

4.1 Reproducing the previously reported fusion advantage

Before re-estimating the fusion effect we checked that any change could be attributed to the protocol and not to our reimplementation. The original selection rule blends raw scores, scans the parameter grid of fusion weights on the evaluation split, and fixes the decision threshold so that the blend calls as many residues epitope residues as the reference predictor does on that same split. Both tuning parameters are therefore optimised on the data the result is reported on. Run through the loaders and metric code used throughout this paper, that rule reproduces the earlier analysis. All 44 entries of its stored weight-sensitivity grid agree to within 4.9e-07, inside the 1e-06 tolerance the check applies and consistent with the precision at which that grid had been rounded. The procedure independently selects the same grid optimum, $w = 0.90$, and the same positive-call count of 1194. The replication script executes this comparison and records its outcome in the artifact (status: reproduced); we do not assert it here. Differences between the

Table 1: The fusion effect under two selection protocols. The test-derived protocol chooses the mixing weight and the decision threshold on the same split it reports; the nested protocol chooses both inside the training folds of a leave-one-sequence-out loop. Intervals are 95% bootstrap intervals over per-antigen differences.

Split	Protocol	Weight	ΔROC-AUC	ΔMCC	Antigens improved
BP3C50ID external test	test-derived	0.90	+0.0018	+0.0115	–
	nested (LOSO)	0.90	-0.0005	-0.0329	7/15
		[0.55, 0.90]	[-0.0033, +0.0021]	[-0.0955, +0.0088]	
IEDB bounded subset	test-derived	0.95	+0.0012	+0.0012	–
	nested (LOSO)	0.95	+0.0008	+0.0014	25/47
		[0.85, 0.95]	[-0.0010, +0.0027]	[-0.0037, +0.0066]	

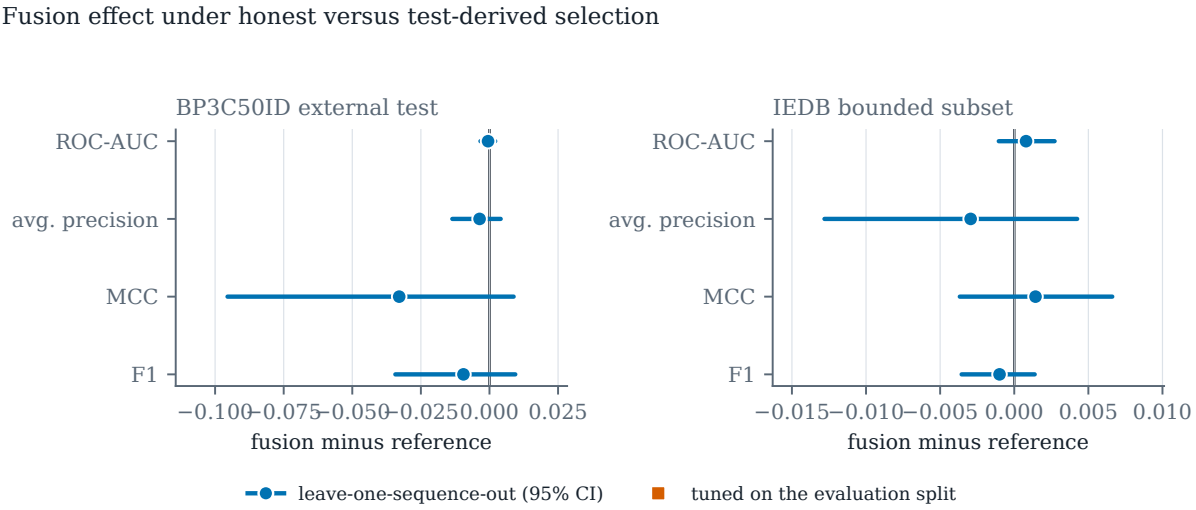


Figure 2: The fusion effect against the reference predictor under two selection protocols. Blue intervals are mean per-antigen differences with 95% bootstrap intervals under leave-one-sequence-out nested selection; orange squares are the corresponding estimates when the weight and threshold are chosen on the evaluation split. On both datasets the nested interval contains zero while the test-derived estimate sits above it.

two protocols below are therefore attributable to how the tuning parameters are chosen, not to a difference in implementation.

4.2 The fusion advantage does not survive nested selection

Table 1 and Figure 2 give both protocols on both datasets, each nested estimate built from one out-of-fold prediction per held-out antigen.

In ROC-AUC, the threshold-independent metric, nested cross-validation leaves a mean per-antigen difference on BP3C50ID against BepiPred-3.0 of -0.0005 [-0.0033, +0.0021], with 7 of 15 antigens improving and a Wilcoxon p of 0.925. MCC, its threshold-dependent counterpart, behaves the same way (-0.0329 [-0.0955, +0.0088], $p = 0.561$).

The larger IEDB subset behaves the same way, with one important difference in what it tests. We have no BepiPred-3.0 predictions for those antigens, so that arm fuses the candidate with our own local-only predictor in place of BepiPred-3.0. That arm poses the same selection question a second time without replicating the same comparison, and we do not read it as independent confirmation of the BepiPred-3.0 result. The mean per-antigen difference in ROC-AUC is +0.0008 [-0.0010, +0.0027], with 25 of 47 antigens improving ($p = 0.485$), and MCC gives +0.0014 [-0.0037, +0.0066]. Every interval on both datasets contains zero. Both the Wilcoxon test and the sign test discard antigens whose paired difference is exactly zero. On BP3C50ID this removes 1 antigens

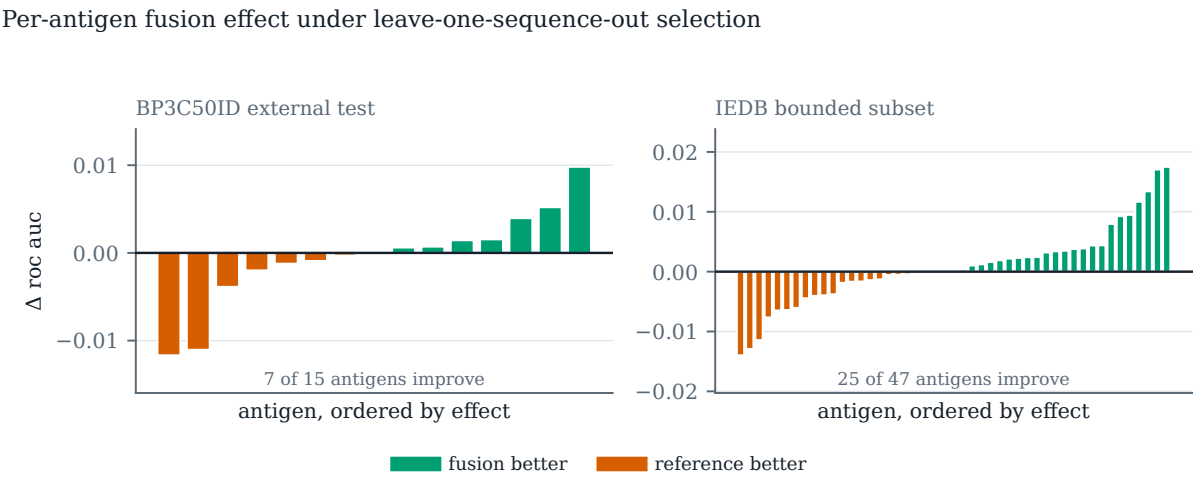


Figure 3: Per-antigen change in ROC-AUC from fusion under nested selection, ordered by effect. The dispersion, particularly on the IEDB subset, is much larger than the mean effect in either direction.

for ROC-AUC and 0 for MCC; on the IEDB subset, 0 and 8. The effective sample size is therefore smaller than the stated number of antigens by that count.

We are careful about what these nulls support. An interval containing zero is not evidence that the effect is zero, and at these sample sizes the paired design cannot resolve effects of the size at issue. With the observed per-antigen dispersion, the smallest difference BP3C50ID could detect with 80% power at a two-sided 5% level is 0.0822 in MCC and 0.0040 in ROC-AUC; on the IEDB subset the corresponding figures are 0.0073 and 0.0027. The MCC effect the test-derived protocol reports, +0.0115, is far below what the nested analysis could have detected. A non-significant nested result was close to guaranteed, so it carries little evidential weight against that specific effect.

What the design does identify is the behaviour of the estimator. Under the test-derived protocol the same BP3C50ID data yield +0.0115 MCC and +0.0018 ROC-AUC in favour of fusion; under nested cross-validation the same quantity on the same data is -0.0329 and -0.0005. The point estimate does not shrink towards zero, it changes sign. This comparison therefore identifies how far model selection bias moves the estimate; it does not identify the underlying effect. The defensible conclusion is that the test-derived estimate is over-optimistic by at least the distance between the two, not that fusion does not help.

Figure 3 shows the per-antigen performance differences behind these means. The dispersion is far larger than the mean in both directions. A null effect over a small number of antigens looks exactly like this, which is why we report the paired tests instead of a pooled number alone.

4.3 Which predictor the procedure prefers

The null effect leaves open what the selection procedure does with its freedom. Figure 4 plots the mixing weight chosen in each training fold of the outer loop. On BP3C50ID the folds settle around a median of 0.90 (range 0.55 to 0.90); on the IEDB subset the median is 0.95 (range 0.85 to 0.95). Both sit close to the upper end of the parameter grid, so the inner loop gives our predictor almost no influence over the blended score.

The selected weights offer an independent route to the same conclusion. The nested evaluation says the blend cannot be distinguished from the reference predictor, and the selected weights say why. Given a free choice, the inner loop declines to use the second predictor. A fusion whose fitted weight approaches one is not a combination of two models but the stronger model with extra machinery attached.

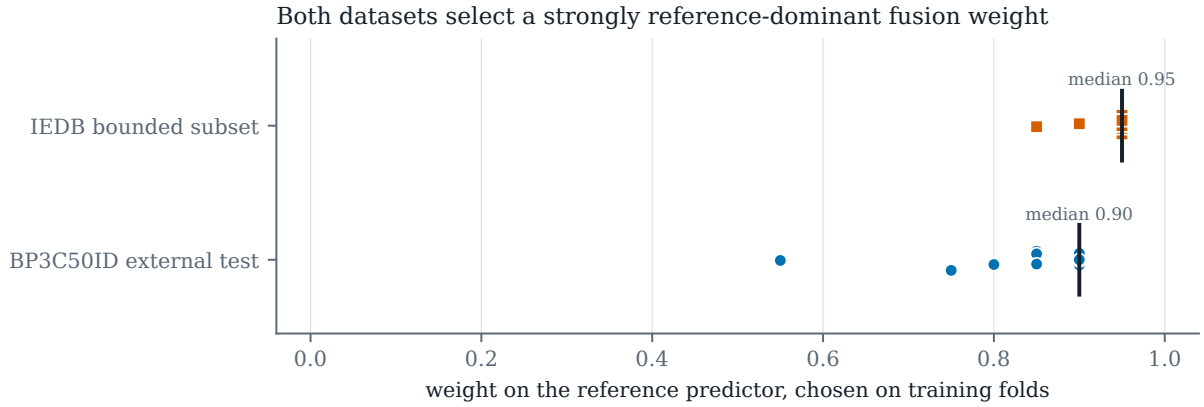


Figure 4: The fusion weight selected in each training fold, by dataset, with medians marked. Both datasets place almost all of the weight on the reference predictor, which is the same conclusion the null effect reaches by a different route.

4.4 How often the published literature selects this way

The result above is a correction to our own unpublished analysis, which bounds how much it matters. Is the practice confined to us? To find out we surveyed published B-cell epitope predictors that combine two or more models, whether by score fusion, ensembles, stacking or consensus voting. Of each meta-predictor we asked whether the combination parameter was chosen using the same data the headline performance is reported on. The survey covers 12 assessable papers and records, for every one, the verbatim sentence the judgement rests on; it is released with the code.

We keep three categories apart. 3 papers are *leakage evident*: the text states that a mixing weight, a voting rule, an ensemble composition or a reported configuration was chosen by inspecting performance on the evaluation data. 6 are *leakage-free evident*: a validation split, the training set, or a rule fixed a priori. 3 are *unreported*: the paper does not say on which data the tuning parameter was chosen. We do not count the third group as leakage. That count observes what the methods section contains, and its size is itself the point — a quarter of the sample does not let a reader check. Among the 9 papers that do state a basis, 33% selected on their own evaluation data.

These counts are not a prevalence estimate and we do not offer them as one. The sample is open-access only, reachable by keyword, and skewed towards recent and prominent work; paywalled methods papers are structurally absent. With 12 papers each count carries a sampling error of one or two. Three things the survey does establish. The selection step we found in our own work is present in the published literature. Among the papers that disclose enough to tell, it is not rare. And a substantial minority of papers disclose too little for the question to be asked at all. The magnitude such a step can reach is the subject of the next section.

4.5 How large the selection step is, and what governs it

The sections above establish that model selection bias moved our estimate and that the practice occurs in print. Neither says how large an effect a reader should expect it to produce. Resampling the same data answers that.

For a given antigen count and parameter grid we draw a subset without replacement, estimate the fused-minus-reference difference on it twice, and record the gap. The first estimate scans the parameter grid on the subset and keeps the best row, which is the protocol a paper follows when it reports a tuned combination parameter without holding anything out. The second applies leave-one-sequence-out selection inside the same subset. Both estimators see exactly the same antigens, so the gap does not come from which data were drawn; it is a difference of protocol, decomposed

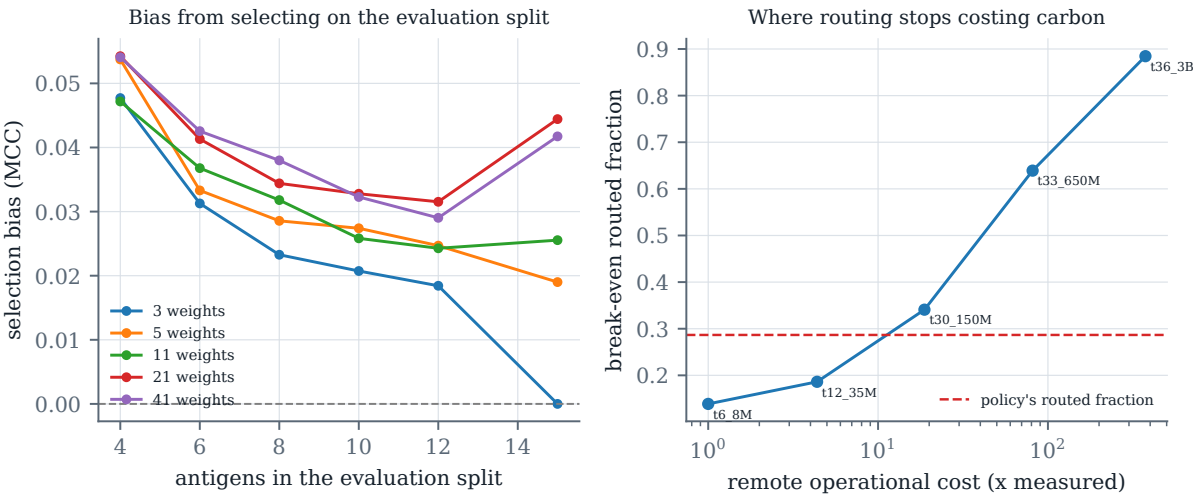


Figure 5: Left: the mean gap between a weight chosen on the evaluation split and one chosen by leave-one-sequence-out selection within it, over 100 draws per cell, by antigen count and weight-grid density. Both estimators see the same antigens in every draw. The rightmost point of each series is the full split, where only one subset exists; it is a single deterministic run rather than a mean, which is why the coarsest grid reaches exactly zero there, and it should not be read as part of the trend. Right: the break-even routed fraction of Equation 10 as the remote tier is scaled, with the policy's own routed fraction marked. The released ESM-2 checkpoints are labelled; the ordering reverses between 35M and 150M. Both panels are modelled from the recorded constants, not metered.

below. Write $\widehat{\Delta}_{\text{test}}(\mathcal{A}; \mathcal{W}_G)$ and $\widehat{\Delta}_{\text{nested}}(\mathcal{A}; \mathcal{W}_G)$ for those two estimates on an antigen subset $\mathcal{A}$, both evaluated over the same grid $\mathcal{W}_G$ of G equally spaced weights. The per-draw gap and the cell mean plotted in Figure 5 are then

$$b(\mathcal{A}; G) \;=\; \widehat{\Delta}_{\text{test}}(\mathcal{A}; \mathcal{W}_G) \;-\; \widehat{\Delta}_{\text{nested}}(\mathcal{A}; \mathcal{W}_G), \qquad \bar{b}(m, G) \;=\; \frac{1}{B_m} \sum_{j=1}^{B_m} b\Big(\mathcal{A}_j^{(m)}; G\Big), \tag{8}$$

In Equation 8, $\mathcal{A}_1^{(m)}, \ldots, \mathcal{A}_{B_m}^{(m)}$ are subsets of size m drawn without replacement from the n antigens of the split, with $B_m = 100$ draws per cell for $m < n$ and $B_m = 1$ for $m = n$, where exactly one subset exists and the cell is a single deterministic evaluation rather than a mean over draws. The two axes of the left panel of Figure 5 are m and G, and the $m = n$ column is excluded from the trend for that reason.

The bias is positive in every resampled cell and large relative to the effects this literature reports (Figure 5, left). On MCC it runs from 0.0542 at 4 antigens to 0.0315 at 12. It grows with the density of the parameter grid. At 12 antigens a 3-point grid yields 0.0184 and a 41-point grid 0.0290. A finer scan simply offers more chances to fit the evaluation split, which is what one would expect. At the grid densities ordinarily used — weight steps of 0.05 are common — the test-derived estimate was the more favourable one in 100% of draws. The full-split cell, the one deterministic point in this series, reproduces the headline of Section 4.2 exactly: at 15 antigens the gap is 0.0444, which is the difference between the +0.0115 that protocol reports and the -0.0329 nested cross-validation returns.

That gap is not wholly a selection effect, and it is worth separating before it is used. The test-derived figure pools residues into one metric, whereas the nested estimate averages per-antigen differences, so the two differ in how they aggregate as well as in where the tuning parameters were fitted. Scoring the test-derived protocol per antigen, exactly as the nested estimator scores it, holds aggregation fixed and isolates selection. Of the 0.0444 MCC headline gap, 0.0388 is selection and 0.0056 is aggregation, so selection accounts for 87% of it.

The comparison worth drawing is between the selection component and the effects it is used to support. Our own earlier analysis had reported a gain of +0.0115 MCC. The measured selection

Table 2: Routing-policy performance on the BP3C50ID external split under antigen-level resampling. Intervals are 95% two-way bootstrap intervals over antigens and seeds (15 antigens, 10 seeds, 2000 resamples), so they carry the uncertainty that matters for a claim about other antigens. Carbon is a model-based estimate, not a metered measurement (Section 3.4).

Policy	MCC	ROC-AUC	Latency (ms)	Raw-sequence share
local-only	0.167 [0.100, 0.229]	0.659 [0.596, 0.712]	0.4 [0.4, 0.5]	0.000 [0.000, 0.000]
edge-only	0.209 [0.150, 0.266]	0.678 [0.621, 0.726]	42.4 [37.2, 48.2]	1.000 [1.000, 1.000]
adaptive-split	0.192 [0.121, 0.255]	0.673 [0.611, 0.724]	13.6 [4.4, 24.0]	0.285 [0.087, 0.507]
carbon-guard	0.165 [0.091, 0.233]	0.658 [0.595, 0.711]	6.6 [1.0, 14.5]	0.133 [0.013, 0.293]
PLM local-only	0.195 [0.124, 0.259]	0.678 [0.614, 0.739]	29.6 [17.2, 51.0]	0.000 [0.000, 0.000]
PLM edge-only	0.209 [0.133, 0.280]	0.691 [0.623, 0.755]	68.3 [54.3, 87.6]	1.000 [1.000, 1.000]
PLM region-constrained	0.219 [0.132, 0.310]	0.690 [0.620, 0.760]	68.3 [54.3, 87.6]	1.000 [1.000, 1.000]
PLM adaptive-split	0.199 [0.124, 0.268]	0.680 [0.615, 0.739]	47.0 [29.3, 70.5]	0.273 [0.107, 0.473]
PLM conformal-guard	0.199 [0.124, 0.268]	0.680 [0.615, 0.739]	47.0 [29.3, 70.5]	0.273 [0.107, 0.473]

Table 3: Paired differences between routing policies under antigen-level resampling, with 95% two-way bootstrap intervals. All four quality metrics are shown for every comparison. An interval containing zero is reported as no detected difference; intervals excluding zero are marked *.

Comparison	ΔMCC	ΔROC-AUC	ΔAP	ΔF1
adaptive-split vs edge-only	-0.017 [-0.066, +0.026]	-0.005 [-0.040, +0.021]	-0.005 [-0.029, +0.027]	+0.002 [-0.030, +0.034]
adaptive-split vs local-only	+0.025 [-0.011, +0.065]	+0.014 [-0.004, +0.038]	+0.015 [-0.001, +0.041]	+0.033* [+0.004, +0.075]
carbon-guard vs adaptive-split	-0.027* [-0.064, -0.001]	-0.015 [-0.038, +0.000]	-0.015* [-0.040, -0.000]	-0.016* [-0.039, -0.001]
PLM region-constrained vs PLM local-only	+0.024 [-0.024, +0.073]	+0.012 [-0.011, +0.033]	+0.018* [+0.003, +0.036]	+0.014 [-0.020, +0.049]
PLM adaptive-split vs PLM edge-only	-0.010 [-0.038, +0.016]	-0.011* [-0.022, -0.001]	-0.014* [-0.026, -0.005]	-0.006 [-0.027, +0.012]

component at this sample size and grid density is several times larger. A tuned fusion result of that magnitude, on a benchmark of this size, therefore carries no information about whether fusion helps, because it sits within what the selection rule alone produces. This is the sense in which the present work generalises beyond one corrected analysis. The bias is not an accident of our predictors; it is a property of scanning a parameter grid on a small evaluation set, and its size can be estimated in advance for any such design.

We are careful about the scope. These magnitudes are specific to these two predictors, whose mean rank agreement across cells is 0.624; two predictors that agree less would leave more room for a scan to exploit, and two that agree more would leave less. What transfers is the construction and the direction, not the numbers.

4.6 What routing buys, and what it costs

Everything above concerns how a configuration is chosen. The rest of this section measures a deployment instead. The question is what routing is worth once the freedom to choose after the fact has been removed.

Table 2 reports every policy on the BP3C50ID external test set across 10 seeds, and Table 3 gives the paired comparisons that the claims below rest on.

Those five comparisons were fixed before the resampling was run: each contrasts a routing policy with the always-local or always-remote baseline it is meant to improve on, within its own

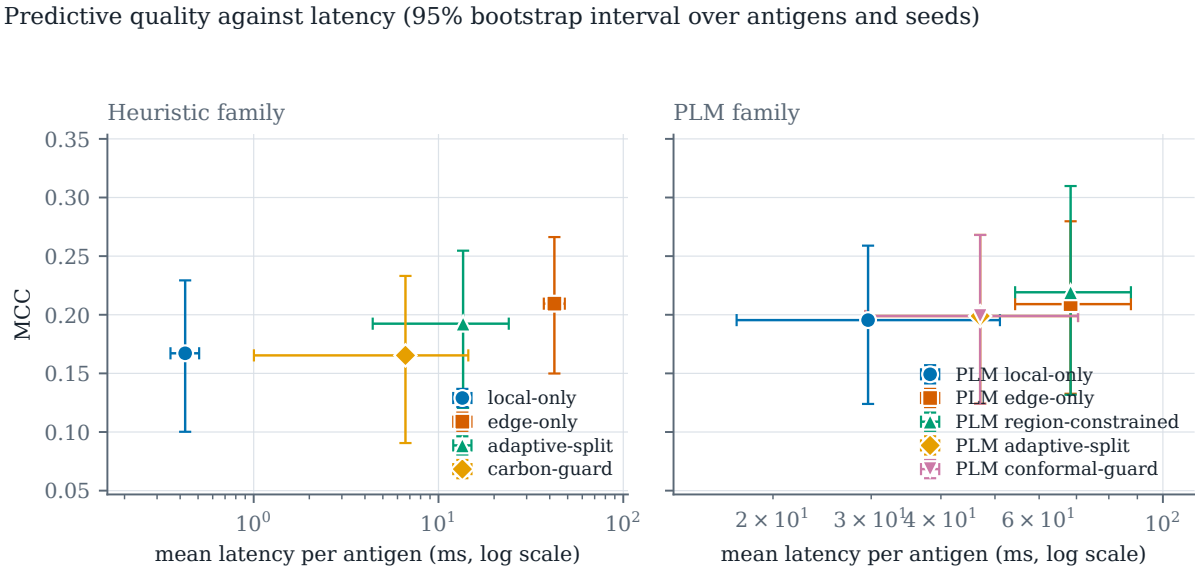


Figure 6: Predictive quality against mean per-antigen latency on the BP3C50ID external split, by policy family. Points are means over 10 seeds; bars are 95% bootstrap intervals on both axes. In the PLM panel the adaptive-split and conformal-guard markers coincide exactly, because the conformal criterion never altered a routing decision on this split.

predictor family. The resampling itself produces intervals for all 36 orderable pairs of policies on each metric, or 144 intervals on the four quality metrics together. We mark intervals that exclude zero in Table 3 as a reading aid for those five rows, and no multiplicity correction is applied. A reader should not treat the marks as a family-wise claim. No family-wise error control is claimed across the 144 intervals, and we do not quantify one: the intervals are built from the same antigens and the same seeds, so they are strongly dependent, and a family-wise rate obtained by assuming independence would be a tidier number with nothing behind it. The absence of such control is exactly why inference is confined to the five pre-specified comparisons and every other interval is reported descriptively. Several of the marked endpoints also sit close enough to zero that the Monte Carlo error of a 2000-resample bootstrap could move them. We draw no conclusion from any single mark. The conclusions below rest on the size and direction of the differences, not on which intervals happen to exclude zero.

Two points about those intervals before the results. They are two-way bootstrap intervals over antigens and seeds (2000 resamples), not intervals over seeds alone. The distinction is not cosmetic. The external test set holds the same 15 antigens in every seed. A seed-only interval therefore measures how much a number moves when the harness is re-run, and says nothing about how it would move on different antigens. On this data seed-only intervals are roughly an order of magnitude narrower: local-only MCC is 0.167 [0.100, 0.229] under antigen resampling against a seed-only width of a hundredth. For two of the policies the seed-only ROC-AUC interval has width zero, because the underlying predictor is deterministic and only the validation split moves. Every interval below is the antigen-level one. We report it this way because the claims are claims about antigens.

The first result is that the latency and disclosure benefits of routing are large and survive this resampling, while the quality questions do not resolve at this sample size. Figure 6 plots predictive quality against mean per-antigen latency for each policy family.

Against always-remote execution, adaptive-split cuts mean latency by −28.8 ms [-37.9, -19.1] and reduces the fraction of antigens whose raw sequence is transmitted by -0.715 [-0.913, -0.493] (Figure 7 shows quality against that fraction). Neither difference could have come out positive: a policy that routes some antigens cannot be slower than one that routes all of them, and cannot transmit more sequences. These intervals therefore say how large the gain is on other antigens, not

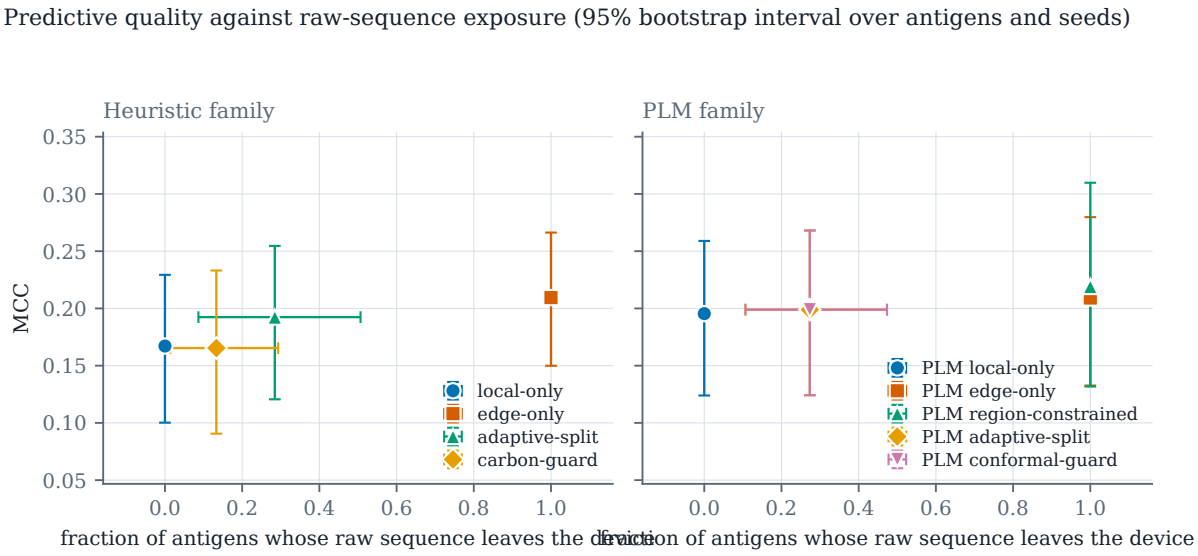


Figure 7: Predictive quality against the fraction of antigens whose raw sequence is transmitted. Policies at zero on the horizontal axis never transmit; edge-only transmits every antigen. Adaptive-split occupies the interior of this trade-off, which always-local and always-remote execution cannot.

whether there is one. On quality, adaptive-split differs by ΔMCC -0.017 [-0.066, +0.026], ΔROC-AUC -0.005 [-0.040, +0.021], ΔAP -0.005 [-0.029, +0.027] and ΔF1 +0.002 [-0.030, +0.034]. All four contain zero.

We want to be careful about what that does and does not license. Those intervals do not demonstrate equivalence. With 15 antigens the interval on ΔMCC spans roughly ±0.05, so a quality loss large enough to matter in practice would not be detected here. The defensible statement is that routing's quality cost relative to always-remote execution is not resolved at this sample size, while its latency and disclosure benefits are.

The comparison against always-local execution tells the same story from the other side. ΔMCC is +0.025 [-0.011, +0.065], ΔROC-AUC +0.014 [-0.004, +0.038] and ΔAP +0.015 [-0.001, +0.041], all containing zero, while ΔF1 is +0.033 [+0.004, +0.075], which does not. One metric of four separates them. Under seed-only resampling the same comparison appears decisive on every metric, with ΔMCC +0.034 [+0.022, +0.045] and a sign test over seeds at $p = 0.0039$. That sign test is a statement about ten re-runs on fifteen fixed antigens, and we no longer rest anything on it.

Two comparisons do survive. The carbon-guard policy is worse than adaptive-split on three of four metrics (ΔMCC -0.027 [-0.064, -0.001]), so its latency and carbon savings are bought with quality. Within the PLM family, always-remote execution beats always-local on ROC-AUC and average precision. PLM routing gives part of that back (ΔROC-AUC -0.011 [-0.022, -0.001] against PLM always-remote). In the family where the remote model is genuinely stronger, routing's quality cost is detectable even here.

Table 4 reports the robustness axes the harness provides. The informative row is the one where the remote tier is unavailable. Every adaptive-versus-local difference is then exactly zero, because routing degrades to local-only behaviour instead of failing. The grid-intensity axes move nothing that matters, and the parameter the carbon results are most sensitive to — the embodied amortisation constant — is not among the available axes (Section 6).

The second result runs against the expectation that motivated the policy. Figure 8 shows modelled carbon per antigen for every policy. Adaptive-split is the *highest* of the nine at 0.160 gCO_2e [0.150, 0.167], above both edge-only (0.139) and local-only (0.120). The paired difference against local-only is +0.040 [+0.031, +0.047] and against edge-only +0.021 [+0.010, +0.028], both intervals excluding zero.

Table 4: Sensitivity of the adaptive-split comparisons to the scenario axes available in the harness. Values are mean ΔMCC over seeds. The grid-intensity axes do not touch predictive quality by construction and are included to show which axes were varied; the remote-tier-unavailable row is the informative one, where routing degrades exactly to always-local behaviour. The dominant carbon parameter, the embodied amortisation constant, is not among the available axes.

Scenario	ΔMCC vs local-only	ΔMCC vs edge-only
baseline	+0.034	-0.010
remote tier unavailable	+0.000	+0.000
network +50 ms	+0.034	-0.010
network +5 ms	+0.034	-0.010
grid intensity ×1.5	+0.034	-0.010
grid intensity ×0.5	+0.034	-0.010

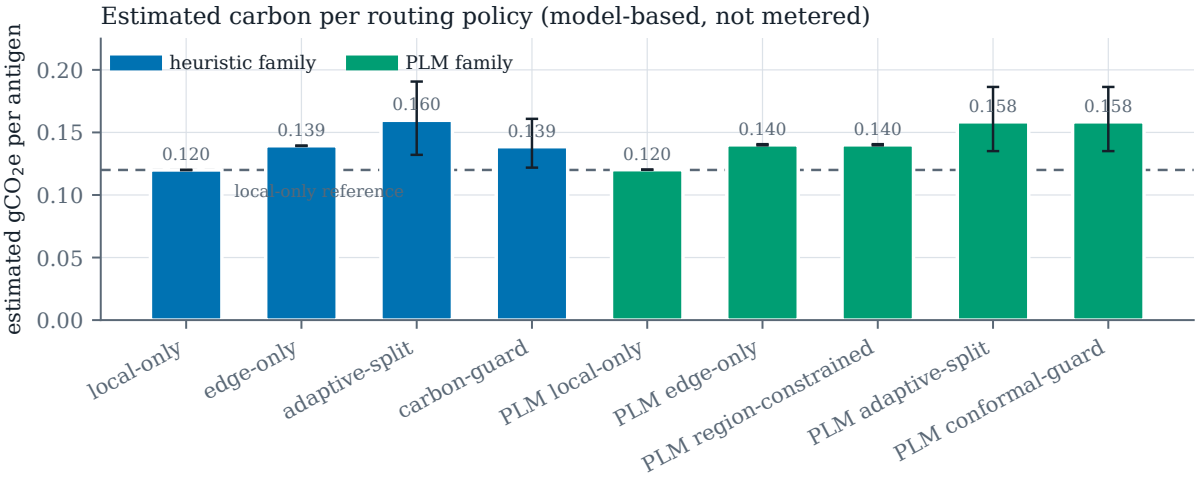


Figure 8: Modelled carbon per antigen by policy, with 95% intervals. These are outputs of the energy model described in Section 3.4, computed from fixed device power, an amortised embodied term and measured latency; they are not metered measurements. Uncertainty-triggered routing is the most carbon-intensive policy here because a routed antigen is charged two device-requests; the ordering is specific to this checkpoint pair and reverses for a larger remote tier, as the text below sets out.

This ordering is not an empirical discovery. Because a routed antigen incurs the local pass that decided to route it and then the remote pass, the modelled carbon of the routing policy is determined by the two single-tier costs and the routed fraction r:

$$C_{\text{adaptive}} = C_{\text{local}} + r\,C_{\text{edge}}. \tag{9}$$

Equation 9 reproduces the measured aggregate to within 0.020% (predicted 0.160 against measured 0.160). Rearranging, routing costs more than always-remote execution exactly when

$$r > 1 - \frac{C_{\text{local}}}{C_{\text{edge}}}, \tag{10}$$

which for this device pair is 0.139. The policy routes 0.287 of antigens, roughly twice that threshold, so it lands on the unfavourable side.

Stated this way the result transfers. Equation 10 says routing saves modelled carbon only where the local tier is cheap relative to the remote tier *and* the routed fraction stays below $1-C_{\text{local}}/C_{\text{edge}}$. The inequality is the offloading energy threshold of Kumar and Lu (2010) written for this cost model. Our system falls on the wrong side of it, because per-residue screening on short sequences makes the gating pass a large fraction of the work it gates. The same inversion has

Table 5: Provenance of each reported cost quantity. Only latency is measured; the energy and carbon figures are derived from fixed device power and a grid-intensity constant, and are therefore model outputs rather than measurements.

Quantity	Provenance
Latency	timed benchmark execution
Energy	estimated from fixed power and latency
Operational carbon	estimated from energy and grid intensity
Embodied carbon	amortized estimate from device metadata

been reported for draft-and-verify decoding cascades (Dutta et al., 2026). What the measurements contribute is the location of the threshold for this task, not the existence of one.

That location, however, is a property of the checkpoint pair, not of the task. It does not survive a larger remote tier. In the recorded cost model the embodied charge dominates: operational cost is 1.7% of the remote total, while the embodied term is levied per request and does not grow with model size. Scaling the remote model therefore grows a term that starts small. Only once that term becomes comparable to the embodied charge does Equation 10 move. Take operational cost to scale with parameter count at fixed sequence length (Figure 5, right). The break-even fraction then rises from 0.139 at the 8M remote tier measured here to 0.186 at 35M, 0.341 at 150M and 0.639 at 650M. Under that scaling assumption the remote cost per request splits into a fixed embodied charge E, levied once per request and independent of model size, and an operational charge that grows with remote parameter count by a factor k normalised so that $k = 1$ at the tier measured here:

$$C_{\text{edge}}(k) \;=\; E + k\,O, \qquad E \;=\; C_{\text{edge}}(1) - O, \tag{11}$$

with $O > 0$ the measured operational charge per remote request. Routing ties always-remote execution when $C_{\text{adaptive}} = C_{\text{edge}}(k)$, which by Equation 9 is $C_{\text{local}} = (1 - r)\,C_{\text{edge}}(k)$: the local pass must cost exactly the unrouted share of a remote call. Substituting Equation 11 and solving for k gives the crossover in closed form,

$$k^{\star} \;=\; \frac{1}{O}\left(\frac{C_{\text{local}}}{1 - r} \;-\; E\right). \tag{12}$$

Because $O > 0$, $C_{\text{edge}}(k)$ increases strictly with k and so does the break-even fraction $r^{\star}(k) = 1 - C_{\text{local}}/C_{\text{edge}}(k)$ of Equation 10; since $r^{\star}(k^{\star}) = r$ by construction, the routing condition $r > r^{\star}(k)$ holds exactly for $k < k^{\star}$ and fails above it. On the recorded constants Equation 12 gives $k^{\star} = 13.0$: the ordering reverses once the remote tier's *operational* charge reaches 13.0× the one measured here, a remote model of roughly 104M parameters at this local scale. Because the embodied term is held fixed in Equation 11 and the operational charge is only 1.7% of the remote total, that is a far smaller change in the cost of a remote call than in k itself.

This matters for the claim, not for the arithmetic. BepiPred-3.0 is the reference predictor a laboratory would actually call remotely, and it is built on ESM-2 650M, several times the 104M at which the ordering turns. Against a remote tier of that size the policy's routed fraction of 0.287 sits well below the break-even. Routing becomes the *less* carbon-intensive choice, not the more. We therefore state the carbon result conditionally. Routing is the most carbon-intensive policy of those compared *for the checkpoint pair measured here*. The finding that transfers is Equation 10, together with the observation that a gate is only worth its carbon when what it gates is substantially more expensive than the gate. Our own pairing, an 8M local model against an 8M remote one, is close to the worst case for that condition. Like the carbon figures it extends, the sensitivity is modelled on the recorded constants, not metered.

The carbon-guard policy, which exists precisely to suppress unprofitable routing, does reduce carbon to 0.139 — essentially edge-only's figure — and cuts latency furthest, to 6.6 ms. It pays for this in quality: against adaptive-split its ΔMCC is -0.033 [-0.046, -0.022], an interval that

excludes zero, and its MCC of 0.173 is statistically indistinguishable from doing no routing at all. On this evaluation the carbon guard does not find a useful middle point; it mostly turns routing off.

Within the PLM family the pattern is similar but compressed, and one component turns out to do nothing. The conformal-guard policy produces per-seed metrics identical to PLM adaptive-split on every metric we record, to the precision stored in the artifacts, because the conformal criterion never changed a routing decision on this split. We report it as a null component instead of omitting it. The region-constrained post-processor, by contrast, is the one PLM addition that helps, improving on PLM local-only by ΔMCC $+0.024$ [$+0.002$, $+0.045$] at no latency cost relative to PLM edge-only.

5 Discussion

5.1 What the fusion result means, and what it does not

Our result is that a convex blend of CARBON-BCE-R with BepiPred-3.0 shows no advantage over BepiPred-3.0 alone, on either of two datasets, once the blend is chosen without reference to the evaluation labels. The scope of that claim matters.

Our result does not show that score fusion is useless in epitope prediction. It bears on one pair of predictors on two splits. One member of that pair is considerably weaker than the other, and the inner scan consistently down-weighted it. A fusion of two comparably strong and genuinely complementary predictors might well behave differently. Nothing here tests that.

We also make no claim that the earlier result was dishonestly produced. The protocol that generated it is easy to arrive at by accident, because a weight sweep serves two different purposes that look the same on the page. Presented as a sensitivity analysis, it answers "how fragile is this result to the weight?" Used to pick the reported row, it answers "what is the best this combination can look like?" The second question's answer is then reported as though it were the first's. The diagnostic is procedural, not statistical. Ask whether the configuration being reported was selected using the labels it is scored against.

What the result does establish is narrower: on data of this kind, the difference between the two protocols is large enough to manufacture a positive finding out of a null one. Here the optimistic protocol had produced a plausible-looking effect with a bootstrap interval attached. Reporting an interval does not protect against model selection bias when the selection happened before the interval was computed. Nested cross-validation is inexpensive: one extra outer loop over antigens, no new data, no new model. We think it should be the default whenever a combination parameter is tuned on a dataset of this size.

5.2 Routing is a disclosure and latency mechanism, not a green one

The case for local–edge routing in a screening pipeline is sound, but it is a different case from the one usually made, and narrower than we expected to be able to make. On our evaluation the mechanism buys two things a laboratory can verify directly, and both survive antigen-level resampling. Screening finishes in roughly a third of the time always-remote execution takes. Most antigens are never transmitted. The second gain is not obtainable by buying a faster remote service.

What it does not buy, on evidence of this size, is a demonstrated quality position. Against always-remote execution the quality difference is not resolved; against always-local execution only one metric of four separates them. We had originally read the same comparisons as decisive, on intervals computed over seeds, and Section 4.6 explains why those were the wrong intervals. A laboratory deciding whether to adopt routing should therefore treat the latency and disclosure gains as established here and the quality question as open. That recommendation satisfies less than the one we set out to make, and it defends better.

What the mechanism does not buy is lower energy, and Equation 10 says precisely when it would. Offloading saves energy only above a ratio of work offloaded to overhead incurred (Kumar and Lu, 2010). Our contribution is to locate that boundary for this task, at a routed fraction of 0.139, and to observe that a policy tuned for predictive quality routes about twice as often as that.

We had initially read that as a negative result about routing. The finding reads better as a statement about the pair being routed between. The boundary is not fixed: it climbs with the cost of a remote call, and crosses the policy's routed fraction once that call's operational charge reaches 13.0 times the one we measured. Our local and remote tiers were both small. Two tiers of similar size sit close to the worst case for a gate, because the gating pass is a large fraction of the work it gates. A laboratory routing to BepiPred-3.0, built on ESM-2 650M, sits several times past that crossing, and for that deployment the green argument recovers. The honest summary is therefore not that routing costs carbon but that a gate earns its carbon only when what it gates is substantially more expensive than the gate. We measured a configuration where it is not.

The practical recommendation follows from the ratio rather than from our sign. Adopt routing when sequence disclosure or turnaround is the binding constraint. On environmental grounds, evaluate Equation 10 for the models actually in use: with a small local screen in front of a large remote predictor it will usually favour routing, and with two models of similar size it will not.

5.3 Relation to prior work

Our benchmark is built on the BP3C50ID split distributed with BepiPred-3.0 (Clifford et al., 2022). The local predictor descends from the feature-based and deep approaches surveyed by Cia et al. (2023) and exemplified in this journal by Attique et al. (2023). We do not advance the state of the art in predictive accuracy and do not claim to. Standalone CARBON-BCE-R remains below BepiPred-3.0 on BP3C50ID, and structure-aware methods such as DiscoTope-3.0 (Høie et al., 2024) occupy ground our sequence-only setting does not reach.

Our evaluation is small beside purpose-built benchmarks for this area such as AsEP (Liu et al., 2024), and we call it a measurement harness rather than a benchmark for that reason. Within the scope of inference routing, however, we are not aware of prior work that routes between a local and a remote conformational epitope predictor and measures the consequences. Federated approaches to protein modelling address a different problem. They distribute training; they do not place inference.

The systems contribution sits alongside the split-computing literature (Matsubara et al., 2022), from which we borrow the routing mechanism instead of inventing it. What we add is the measurement of that mechanism against the objectives a wet-lab screening workflow actually has. That includes the counting measure of raw-sequence disclosure, which accuracy-oriented formulations do not express, and the observation that the energy argument inverts at this scale.

The methodological contribution is an instance of the problem Kapoor and Narayanan (2023) document across applied machine-learning fields, reported from the inside: we are correcting our own earlier result rather than auditing someone else's. We think instances of this kind are useful precisely because the published record contains few of them.

6 Limitations

Several constraints bound what can be concluded from this study.

Carbon is modelled, never metered. Every energy and carbon figure in this paper is computed from a fixed device power constant, a measured latency and a grid-intensity constant (Section 3.4, Table 5). No wall-power measurement was taken at any point. The carbon comparisons between policies are therefore comparisons of modelled quantities that are proportional to measured latency; they support statements about which policy does more work, and they do not

support any statement about emissions of a deployed system. A metered replication on instrumented hardware is the obvious next step and is not something this study attempted.

The carbon figures rest on two utilisation constants. The energy model reproduces, but only once the utilisation fractions are supplied by hand. Across all 47 antigens of the second dataset, recomputing each per-antigen total from the recorded device configuration, the measured latencies and the utilisation fractions of Section 3.4 matches the stored values to within 0e+00 gCO_2e. The fractions are not in the recorded configuration, so the stored totals cannot be rebuilt from that configuration alone, and the released decomposition artifact says so. The fractions themselves, however, are literals in the benchmark code rather than entries in any configuration file, and they dominate the result: the embodied term is almost the whole of each per-request total.

That matters for how far the carbon conclusion travels. Of 5 alternative settings we evaluated, the ordering survives in all but 1. A device pair with the local tier at 0.10 utilisation and the remote tier at 0.90 would put the break-even routed fraction at 0.690, above the rate our policy actually routes at. Routing would then be the greener option. Our finding therefore holds for the hardware pair modelled here and for most reasonable variations on it. It remains a statement about a cost model whose dominant parameters deserve to be measured rather than assumed. A metered replication remains the right next step.

The remote tier is co-located, not remote. The latency figures attribute a transport cost to routed antigens, but the benchmark executes both tiers on one machine, and the two-node transport evidence in the artifact is an HTTP probe, not a deployed remote model server. Real deployments would add network variance, queueing and failure modes that this harness does not model. For latency and for transmitted volume these would, if anything, favour routing, not count against it, since always-remote execution pays the network on every antigen and routing pays it on roughly a quarter. What a real deployment would add against routing is of a different kind: a dependency on the remote tier for exactly the antigens the local model found hardest, and the failure and retry behaviour that comes with it. We measure none of that.

The primary benchmark is small, and that bounds nearly everything. BP3C50ID's external split has 15 antigens. Resampling on 15 units gives intervals wide enough that most quality comparisons in this paper are not resolved, and every such comparison should be read as "no effect detectable at this sample size" rather than as "no effect". This applies to the routing results as much as to the fusion analysis: the interval on the adaptive-split versus always-remote MCC difference spans roughly ± 0.05, which would not exclude a quality loss a practitioner would care about. For the fusion analysis the IEDB subset at 47 antigens carries more weight on ROC-AUC and average precision, where it has no tied folds. On the threshold-dependent metrics its informative-fold count is closer to BP3C50ID's and we do not claim it is better powered there.

The language-model tier is the smallest available checkpoint. The PLM results use `esm2_t6_8M_UR50D`, chosen for the hardware at hand. Conclusions about the PLM family are conclusions about an 8M-parameter frozen encoder and should not be read as characterising what language-model features can contribute to this task.

One routing signal, one validation design. Routing decisions come from mean per-residue local uncertainty, selected on a single grouped validation split per seed. Repeated grouped resampling of the validation split would give a better picture of how stable the selected thresholds are; we vary the seed, which moves the split, but we do not separate the two sources of variation.

The IEDB subset's sampling rule is not recorded. Its source and identity filter are established (Section 3.1), but how the 47 test and 205 training antigens were drawn from the 3560 available is not captured in any script we could find. The selection is therefore reproducible in its inputs but not in its sampling. We state this rather than reconstruct a rule after the fact, since a plausible-looking rule invented now would be indistinguishable in the write-up from one that was actually followed, which is the failure this paper is about. What we can do is pin down the set: the released manifest lists every test antigen by accession, so the evaluation subset is identified exactly even though the draw that produced it is not. The consequence for the reader is that the IEDB arm should be treated as a second dataset of opportunity rather than as a designed replication, and we do not rest the paper's conclusions on it.

Declarations

Data and code availability. The benchmark harness, the analysis modules that produce every table and figure in this paper, and the raw per-seed result artifacts are available at `https://github.com/Anandharajan/edgeconbce`. A single command (`regenerate.sh`) rebuilds the generated-number file, all tables, all figures and this manuscript from the stored results. The repository is archived on Zenodo at acceptance and the DOI will replace this URL in the published version.

The BP3C50ID training and external test splits, and the `IEDBSeqsNotSharedAt20ID.fasta` collection from which the second evaluation set is drawn, are redistributed by the Technical University of Denmark alongside BepiPred-3.0. We do not republish them, and the repository contains no antigen sequences from either source. The BepiPred-3.0 predictions used as the reference were obtained from the official service.

The rule by which the 47 test and 205 training antigens were drawn from that collection was not recorded when the subset was made, and we have not been able to recover it. We therefore release the identity of the evaluation subset instead (`data/iedb_subset_manifest.csv`), which lists every test antigen by accession, so that the set itself can be reconstructed from the source collection even though the draw cannot be repeated. The 205 training antigens are counted in the released artifacts but not identified, and we do not reconstruct them after the fact.

Ethics. This study uses only published protein sequence data. No human participants, human tissue, animal subjects or personally identifying information were involved, and no ethics approval was therefore required.

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